KERALA STATE COUNCIL FOR SCIENCE, TECHNOLOGY & ENVIRONMENT (KSCSTE)

APPLICATION FORMAT FOR STUDENT PROJECT SCHEME

1. **Name(s) of Student Investigator(s):**

1. Sreedevi R P

2. Vishnu V Nair

3. Abhi Raj

2. **Semester and Year of Study:**

7th Semester, 4th Year B.Tech Computer Engineering

3. **Address of the College:**

College of Engineering Chengannur, Chengannur (P.O), Alappuzha, Kerala, PIN 689121

4. **Category of the College (Govt./Aided/Private):**

Aided

5**. Residential Address with Telephone Number & Email:**

Kollantharayil, Keezhcherimell, Chengannur, Chengannur

Phone: 6282925344

Email: abhirajcsecec@gmail.com

6**. Title of the Project Proposal:**

CyberShield AI: A Web-Based Real-Time Cybersecurity Monitoring Dashboard with AI-Assisted Threat Analysis.

7**. Achievement(s) of the Student(s) in Science-Related Activities:**

Authors of academic literature survey papers on Machine Learning in Cybersecurity for CSD415 Project Phase 1.

Active core student members in College of Engineering Chengannur Innovation and Entrepreneurship Development Centre (IEDC)

Successful development and deployment of open-source full-stack and machine learning software solutions.

8. **Name of Teacher Supervisor / Mentor:**

Varsha C Mohan

9. **Objectives of the Project:**

To develop a web-based cybersecurity monitoring dashboard that aggregates security telemetry and provides real-time alerts.

To implement on-demand phishing-URL and malware scanning capabilities.

To integrate the Google Gemini API to generate human-readable, expert explanations for detected threats to assist small IT teams and Security Operations Centres (SOCs).

To bridge the gap between offline threat detection accuracy and actionable, real-time response tooling.

10. **A Brief Description About the Project (Max. 300 words):**

CyberShield AI is a layered client-server web application designed to empower small IT teams and Security Operations centres (SOCs) with accessible, real-time threat detection. Built with a React/Vite frontend and an Express.js/TypeScript backend, the dashboard aggregates security metrics and provides on-demand scanning for phishing URLs and malware. Unlike standalone detection models, CyberShield AI integrates the Google Gemini API to analyse flagged threats and automatically generate human-readable, expert explanations, making complex security alerts easily understandable. The system features real-time network monitoring, threat-intelligence feed integration, and a simulated in-memory data store that refreshes every 10 seconds. Furthermore, detected anomalies trigger automated, severity-categorized alerts pushed directly to configurable Discord or Slack webhooks, ensuring immediate visibility and actionability for security teams.

11. **Relevance of the Project:**

Current research in cybersecurity often treats phishing, malware detection, and human-centred security in isolation, resulting in high detection accuracy but poor real-time

applicability for human analysts. CyberShield AI directly addresses this gap. By combining heuristic scanning, live dashboard visualization, and LLM-assisted explanations, the project ensures that raw security data is immediately translated into understandable, actionable insights. This is critical for small organizations that lack the budget for enterprise-scale SOC platforms but still require robust, explainable security monitoring.

12**. Time Required to Complete the Project:**

6 Months

13. **Whether Assistance from Any External Agency/Institution is Required to Complete the Project:**

No.

14. **Estimate Showing the Expenditure Involved in the Project:**

Consumables / Cloud Infrastructure (AWS/GCP, Domain): 4,000

API Costs (Google Gemini / VirusTotal Enterprise): 3,500

Contingencies / Report Preparation: 2,500

Total Expected Cost: 10,000

DECLARATION BY THE STUDENT(S)

We hereby declare that the details furnished above are true and correct to the best of our knowledge.

Signatures of the Student Investigators:

1. _____ (Sreedevi R P)

2. _____ (Vishnu V Nair)

3. _____ (Abhi Raj)

ENDORSEMENT BY THE TEACHER SUPERVISOR

I have scrutinized the project proposal titled "CyberShield AI" and I recommend the same for financial assistance under the Student Project Scheme of KSCSTE.

Signature of the Teacher Supervisor: _____

Name & Designation: Varsha C Mohan (Assistant Professor, Dept of Computer Engineering)

ENDORSEMENT BY THE HEAD OF THE INSTITUTION

This is to certify that the above-mentioned students are Bonafide students of this institution. If the project is sanctioned, the institution will provide basic infrastructure and facilities to complete the project.

Signature of the Principal / Head of the Institution: _____

Name & Designation: Prof. (Dr.) Hari. V S, (Principal, College of Engineering Chengannur)

Date:

Office Seal: